Edward Silva

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Experience

Software Engineering Intern, Kratos Defense – Colorado Springs, CO — June – Aug 2025

- Achieved 1.6x execution speedup by optimizing legacy DSP algorithms in C++ through code refactoring and performance analysis, reducing computational overhead for real-time signal processing applications
- Improved system throughput by developing and implementing SIMD-optimized mathematical algorithms using vectorized operations for parallel data processing
- Researched and demonstrated an improved approach to coding a FIR filter, presenting positive findings and performance gains to the team for adoption in future projects
- Reduced debugging time and improved system maintainability for development teams by designing and deploying a comprehensive logging framework with configurable severity levels and error tracking

Projects

Non-Linear State Estimation for the EMBR Damper, MATLAB, Simulink — Spring 2026

- Architected a non-linear state estimation strategy for an active electromagnetic suspension system, modeling the dynamic interactions of a custom magnetorheological (MR) fluid damper, MOSFET power regulation, and a multi-sensor measurement array
- Formulated continuous and discrete-time non-linear state-space models in MATLAB and Simulink, simulating system kinematics to validate damping performance against harmonic, stochastic, and discrete terrain disturbances
- Implemented Extended (EKF) and Unscented (UKF) Kalman Filters for unmeasured state estimation, applying the UKF to eliminate analytical linearization errors and tightly bound state error covariance during highly non-linear fluid yield transitions

Autonomous Maze-Navigating Robot, MATLAB, Simulink, Python, Arduino — Fall 2025

- Co-developed an autonomous maze-navigating robot within a 4-person team, integrating computer vision, embedded control, and I2C communication to achieve the second-fastest course completion time in the cohort
- Built the computer vision subsystem using Python and OpenCV with ArUco marker detection to extract real-time distance and angle metrics for navigation
- Designed a multithreaded vision pipeline processing 640x480 video feeds with HSV color masking to identify directional markers for pathfinding logic
- Interfaced the vision subsystem with motor control via I2C bus, transmitting telemetry data to an Arduino while updating a real-time RGB LCD display

Skills

Programming Languages: C++, Python, MATLAB, Java, Verilog, RISC-V Assembly, Bash
Hardware: Arduino, Raspberry Pi, Digital Circuits, Embedded Systems, Microcontrollers, Circuit Design
Software: Git, Simulink, SSH

Education

Colorado School of Mines — May 2026
BS Electrical Engineering, Minor in Computer Science
Relevant Coursework: Signals and Systems I/II, Algorithms, Digital Signal Processing, Estimation Theory & Kalman Filtering, Modern Control Design, Embedded Systems, Computer Architecture, Software Engineering